

Section 1.4: Recent Developments in ICT

ICT is constantly developing and expanding as new technologies become available. In this section we will look at some of the more recent developments in ICT.

Learning objectives of this section:

- # Be able to **describe** and discuss some of the recent developments within ICT.

Some recent developments.....

The recent developments in ICT that we will discuss below will focus on two areas:

- **Cloud Computing**
- **E-books.**

1 Cloud Computing

Cloud computing is where users **store** their **documents, programs** and **data** on the **Internet** rather than on their own computers.

As long as the user has access to an Internet connection, they can **create, edit, and share ICT files** from almost **any location**.

Because the data and programs are **stored remotely**, on the Internet, the user will save space on their computer's storage drives.

Some uses of Cloud Computing

Streaming music

- ➔ Music is **stored remotely on an Internet server** and accessed via a website.
- ➔ Users can log onto the website and then listen to the music using '**Internet Streaming**'.

NOTE:

Streaming music is where users play the audio directly from the Internet server where it is stored.



Key Words:

Cloud, E-book, E-book reader, Remote, Share, Streaming, Google Docs

Examples:



Cloud computing allows users to store and access documents and programs from the Internet.
(click image to zoom)

- ➔ The music file is **not downloaded onto the users computer**.
- ➔ Streaming music is a little bit like listening to the radio where you listen to the music but do not actually own a copy. The only difference is that, with streaming, you get to pick which songs you listen to.
- ➔ Popular music streaming websites include:
 - [Spotify](#)
 - [Jango](#)
 - [Pandora](#)

Google Docs

- ➔ Google docs offers users the ability to use **free software** such as **spreadsheets, word processors, presentation builders** and **drawing programs**.
- ➔ This software is stored on the Internet (in the Cloud) so users don't have to install it to their computers.
- ➔ Any documents created within Google Docs are also **stored remotely** in the cloud. This means that the documents can be **accessed from any location** as long as the user has an Internet connection.
- ➔ You can see how Google Docs works for yourself by clicking on the link below:
 - docs.google.com

NOTE:

Google Docs is very useful for group projects because all of the members of the group can work on the document without needing to be in the same room.



There are lots of online sites that allow you to stream music and listen to it on a computing device.
(click image to zoom)



Google docs allows you to use online programs to create a range of documents. These documents are stored and accessed (remotely) on the Internet.
(click image to zoom)

Advantages and Disadvantages of Cloud Computing

Advantages of cloud	Disadvantages of cloud
Lower costs: Many programs in the cloud are free to use.	Security risks: Data stored online is always vulnerable to hackers and viruses.
Increased accessibility: Programs and documents are accessible no matter where you are as long as you have an Internet connection.	Decreased accessibility: If you do not have access to the Internet (your connection is down) then you cannot access your documents and programs.
Time saving: No need to spend time installing software onto your computer as you access it directly from the cloud.	Quality of software: Cloud programs (especially the free ones) don't usually have all of the features of a full version of the software.
Increased storage space: Programs and documents are stored in the cloud so this frees up storage space on your computer.	For example: cloud's version of a word processor may not allow you to edit a document as well as Microsoft Word does.
Backups: Documents in the cloud are automatically backed up for you so you don't have to remember to do it yourself.	Potential increased costs: Some cloud providers only allow you to use their programs after you have paid a subscription.

Sharing of data:
Documents can be shared very easily which is useful for group work.

2 E-books and Online-newspapers (electronic books and newspapers)

- # E-books and online newspapers are **digital versions of traditional printed publications**.
- # E-books and newspapers are readable on a variety of electronic devices such as:

- **Desktop PC's**
- **Laptops**
- **I-Pads**
- **Mobile Phones**
- **Amazon Kindles**

Amazon Kindles are specifically designed to allow users the ability to read E-books. These types of devices are known as '**e-book readers**' or '**e-readers**'.

- # E-books have already been created to **replace many of old printed books**. There are many advantages of doing this including **saving paper** and **storage space**.

NOTE:
Some new publications are created only as e-books with no printed versions at all.

- # E-books and newspapers are usually **downloaded from the Internet** onto the computer or e-reader.

Electronic books and newspapers tend to be **cheaper to buy** than their printed counterparts.

Examples:



Amazon Kindle's are the most well known example of an e-book reader.



An online newspaper being read on an iPad.

Advantages and Disadvantages of E-books and Online-newspapers

Advantages	Disadvantages
Storage benefits: Thousands of digital books can be stored on one computing device or e-book reader. The same number of printed paper books would require a very large room in which to store them.	Health and Safety: Computer screens (and some e-book readers) can reflect light which causes glare. This can cause headaches and eye strain. Printed pages do not have this problem.

Paperless: Digital books do not need to be printed onto paper. This is much more environmentally friendly.	Piracy: E-books are digital and so are very easy to copy and share.
Instant access to your book or newspaper: E-books/newspapers can be downloaded from the Internet almost instantly. If you want access to a printed book, you need to travel to the store to buy it or order online and then wait a few days for delivery.	Usability: E-books are hard to search through to find specific pages. The pages of printed books can be flipped through quickly to find the page you want.
Cheaper to buy: E-books/newspapers generally cost less than printed publications.	High initial costs: Before you can read a digital e-book or newspaper you need either a computer or an e-book reader. These are not cheap.



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